

Project - 1

Part - A theory & definition

① Define the following with examples from dataset

* Types of data numerical vs categorical

Ans:- numerical:-

it is also called quantitative data, it has two types of numeric data

i) discrete:- it contain without floating point values

Ex ID, age of household head,
family size

ii) continuous:- it contain floating point values

Ex average age of household head

categorical:-

it is also known as qualitative data, it has two types of categorical data

i) ordinal:- it contain categorical data which have specific order

Ex difficulty level (easy/medium/hard)
education level (hard)

ii) nominal:- it contain categorical data which don't have specific order.

Ex owns house

* Types of Statistics: descriptive Vs Inferential

* descriptive:- it is used to summarize and describe the data

key concepts include mean, median and mode etc

* Inferential:- it is used to draw conclusion and make prediction for future analysis.

key concepts include hypothesis testing, confidence interval, probability etc

* What is descriptive statistics?

Ans:- it is used to summarize and describe the data, it include measure of central tendency, measure of dispersion etc

② Explain the difference between:

* Mean, median, mode

Ans:-

Mean is the average value of dataset

Formula:- $\text{mean} = \frac{\text{Sum of all values}}{\text{total no. of values}}$

Median is the middle value of dataset

Formula:- ~~me~~ if length of dataset is ~~odd~~ ^{even} then

$\text{median} = \frac{\text{data}[\text{mid}-1] + \text{data}[\text{mid}]}{2}$

if your dataset length is ~~even~~^{odd}
 then $\text{medium} = \text{data}[\text{mid}]$

mode is the most repeated value of dataset

Formula:- if dataset's data count is 1
 then it return 1

otherwise

it return maximum count's value

* Range, variance, standard deviation

Ans:-

Range is the difference between maximum and minimum value.

Formula:- $\text{Range} = \text{maximum value} -$

minimum value

variance is the squared difference of the data point and mean

Formula:- $\text{variance} = \frac{\sum (x_i - \bar{x})^2}{n-1}$ (For sample)

$\frac{\sum (x_i - \bar{x})^2}{n}$ (For population)

Standard deviation show how each data point spread around the mean

Formula:- Standard deviation = $\sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$

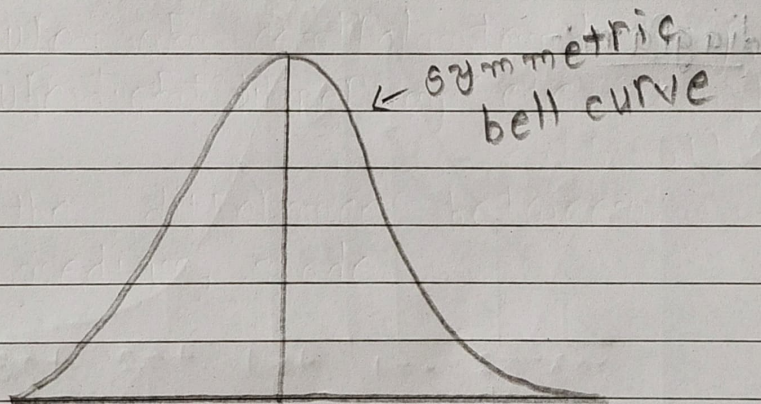
③ Explain the Following term with neat diagram with its Formula:

* Gaussian distribution:-

it is one type of dist., when your data make this distribution, your data doesn't contain outliers

Formula:- $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

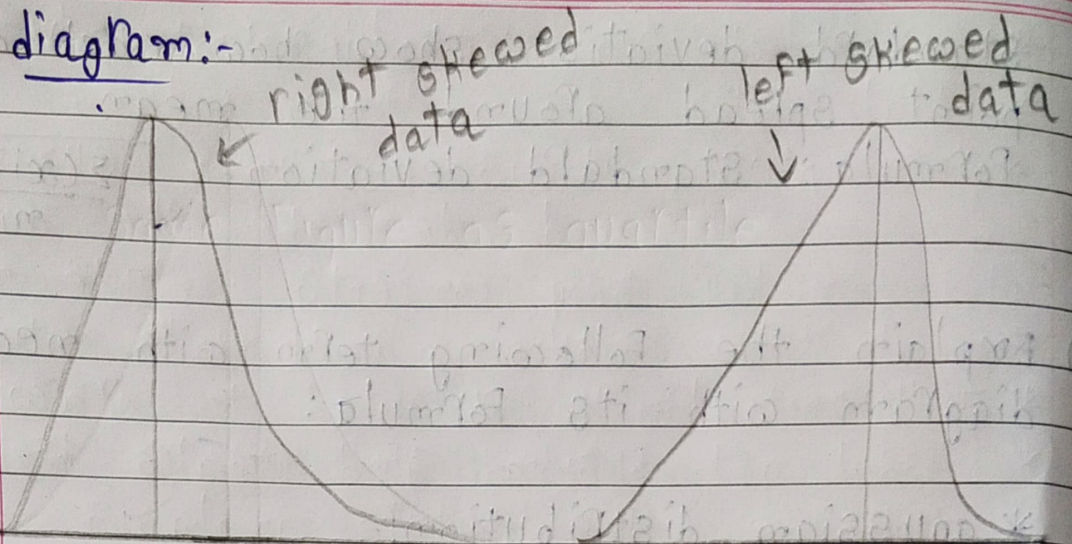
Diagram:-



* log normal distribution:-

it is opposite side of normal distribution, when your data make this distribution, which mean it contain outliers

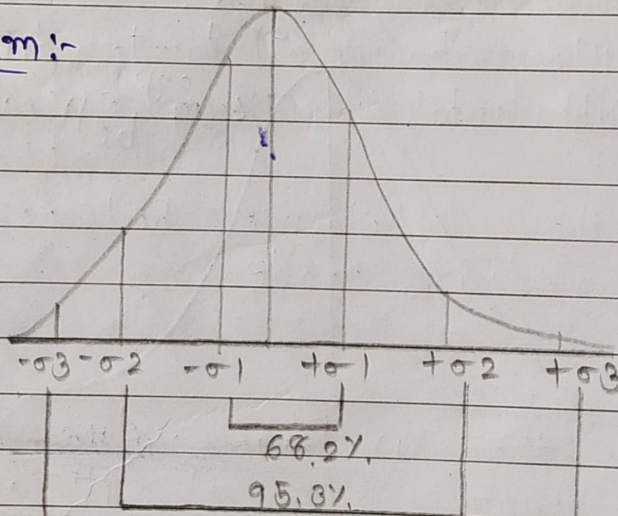
Diagram:-



* 3 sigma rule:-

it is used to make histogram

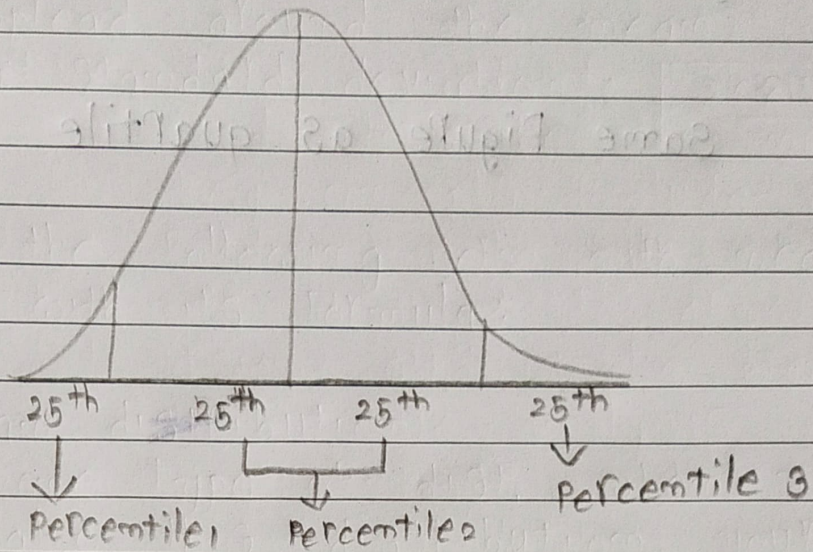
Diagram:-



* percentile:-

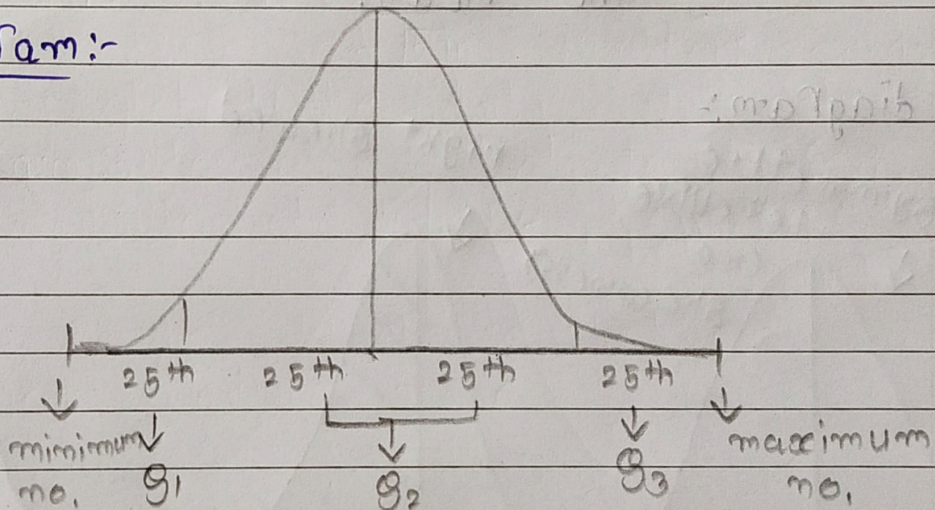
percentile percentage show you're above ^{from} your given percentile percent - age

diagram:-



* quartile:- it give you specified percentage of data from given dataset

diagram:-



* Five number summary:-

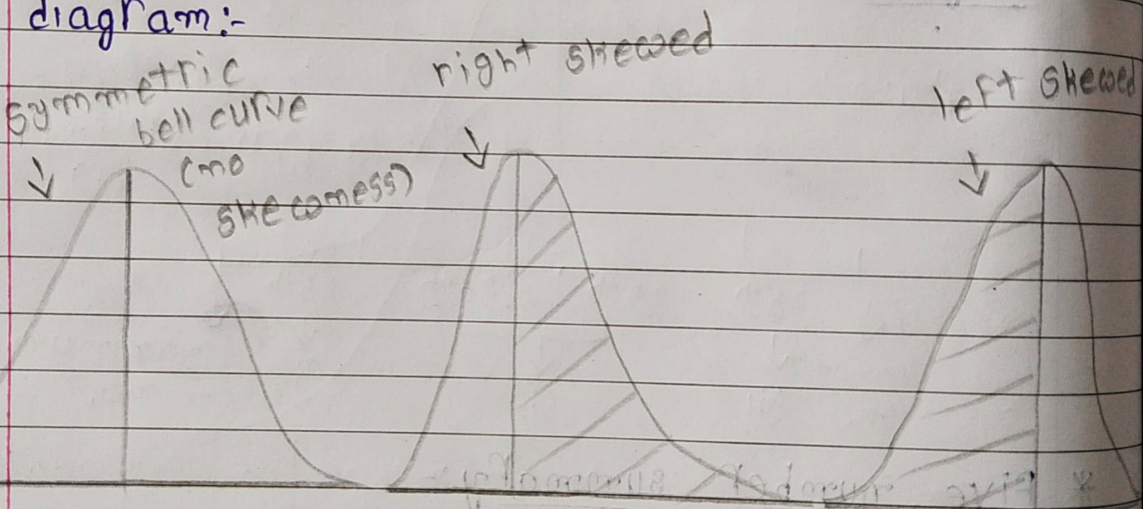
this rule is used to boxplot.

diagram:-

Same Figure as quartile

* skewness:- it measure asymmetry of data distribution around the mean.

diagram:-



* kurtosis:- it measure how many outliers your dataset contain based on type of kurtosis.

There are three types:-

- * leptokurtosis
- * mesokurtosis
- * platykurtosis

diagram:-

